ICS 433 Operating Systems Lab, Phase 2 Progress Report

Virtual Memory Management Simulator

Replicating "FIFO can be Better than LRU" (HotOS 2023) for a shared-server page cache

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1. Project recap

We are building a simulator for a multi-process virtual memory system. Processes share one global pool of physical frames, and FIFO, LRU or Clock decides which page to evict. The simulator tracks page tables, dirty pages, per-process working sets and fault rates. It also raises a simple thrashing warning.

Changes since the proposal: following the instructor's guidance, (1) the simulator is now written in C, and Python is used only to run experiments and draw charts; (2) the project is anchored on a recent paper that we replicate; and (3) it targets one concrete use case.

Reference paper: J. Yang, Z. Qiu, Y. Zhang, Y. Yue, K. V. Rashmi, "FIFO can be Better than LRU: the Power of Lazy Promotion and Quick Demotion", HotOS 2023 [1]. On 5,307 block and web cache traces, it finds that FIFO-Reinsertion (the same algorithm as 1-bit Clock and Second Chance) has a lower miss ratio than LRU, against common belief. In its footnote 2, the paper guesses that this may not hold for virtual-memory workloads, where "LRU can better adapt to sudden working set changes". The authors did not test this.

Use case: the page cache of a shared server, where several processes compete for a fixed pool of frames. Is the cheap Clock family good enough, or does LRU win when processes change their working sets?

2. What has been implemented

About 75% of the system is done. All core modules work and are tested.

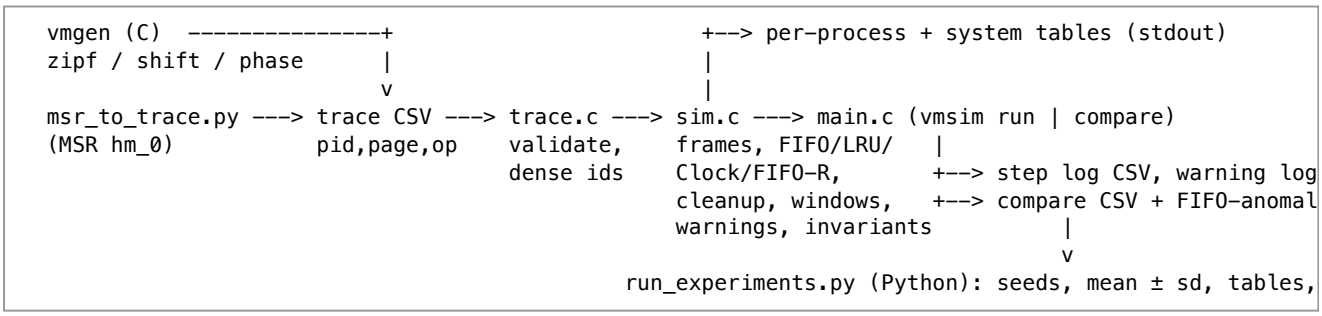
Module	Status	Files
Strict CSV trace loader (exact header, field count, positive PID, non-negative page, R/W only, no blank rows, line-numbered errors)	Done	`src/trace.c`
Simulation engine: global frame pool, lowest free frame first, (pid, page) identity, dirty bits, page-ins, eviction write-backs, completion flushes, process cleanup	Done	`src/sim.c`
Policies: FIFO, LRU, Clock (proposal rule) and FIFO-Reinsertion (the paper's variant)	Done	`src/sim.c`
Working-set windows (last 20 references per process), total demand D, thrashing warnings with an ON/OFF log	Done	`src/sim.c`
Executable invariants (`sim_check`), checked after every reference in the tests or with `vmsim run -c`	Done	`src/sim.c`
Command line: `vmsim run` (per-process tables, logs) and `vmsim compare` (all policies, many frame counts, FIFO-anomaly check)	Done	`src/main.c`
Seeded workload generator: Zipf, shifting working sets, and the proposal's phase trace	Done	`src/gen.c`
Real trace: MSR Cambridge `hm_0` (3,993,316 references), converted from the paper authors' public copy	Done	`tools/msr_to_trace.py`

Unit tests: every case in the proposal's test table, plus the feedback items (529 checks, all passing)	Done	`tests/test_sim.c`
Experiment runner: seeds, tables, charts	Done	`analysis/run_experiments.py`
More real traces, design document, final report and demo	Remaining	Section 7

Response to the proposal feedback

Feedback	What we did
1. Keep the model rules as executable invariants	`sim_check()` verifies: hits + faults = references; no (pid, page) is resident twice; each frame has one owner; occupancy ≤ F; finished processes own no frames; per-process counts sum to the totals; D equals the sum of active working sets; the FIFO/LRU queue length equals the number of resident pages. The tests call it after every reference, including on 20 random multi-process traces × 4 policies × 5 frame counts.
2. The warning is a heuristic; compare onset across working-set sizes	Labelled "heuristic" in the code, the CLI output and this report. Onset was measured for middle-phase working sets of 3, 4, 5, 6, 8, 12 and 16 pages (Fig. 4).
3. Independent random seeds; summarise variability	Synthetic workloads use seeds 11, 23 and 47. Tables report the mean ± sample standard deviation.
4. Cleanup/reuse test with the Clock hand and per-process totals	Test `test_cleanup_and_reuse`: PID 1 finishes and flushes one dirty page. Its freed frames 0 and 2 are reused in order, and the Clock hand is checked after every step (0 → 1 → 0 → 1). Per-process totals are checked, and the FIFO/LRU queues are confirmed empty.
5. Separate page-in, write-back and flush counts	Separate columns everywhere: `page-ins`, `evict-wb` and `flushes` in the CLI; `page_ins`, `evict_writebacks` and `completion_flushes` in every CSV and table.

3. System architecture



- Dense ids: the loader maps every (pid, page) pair to a small integer, so the simulator uses plain arrays (page table `where[key]` = frame). One run over the 4-million-reference MSR trace takes about 2 s.
- One queue for FIFO and LRU: a doubly linked list of frames. Loads go to the tail and the victim is the head. LRU also moves a page to the tail on every hit. All operations are O(1).
- Clock and FIFO-R: a reference bit per frame plus a hand. They differ in exactly one line: whether a page load sets the bit (Clock, our proposal's rule) or leaves it at 0 (FIFO-Reinsertion, the paper's rule).

4. Results so far

Workloads: Zipf (4 processes, 1,000 pages each, α = 0.9, 30% writes; a skewed workload like the paper's block and web traces), the real MSR hm_0 block trace [2], and shift (the same as Zipf, but each process moves its

whole working set to new pages 10 times; a VM-like workload with sudden phase changes). The cache sizes are the paper's 0.1% and 10% of distinct pages, plus 1%. The metric is the paper's: the reduction of the miss (fault) ratio relative to FIFO.

Validation against the paper: on MSR hm_0 at 0.1% of pages, our LRU fault ratio is 0.5263. The paper's Table 2 reports 0.5263, an exact match, so our simulator and trace conversion reproduce the paper's setup.

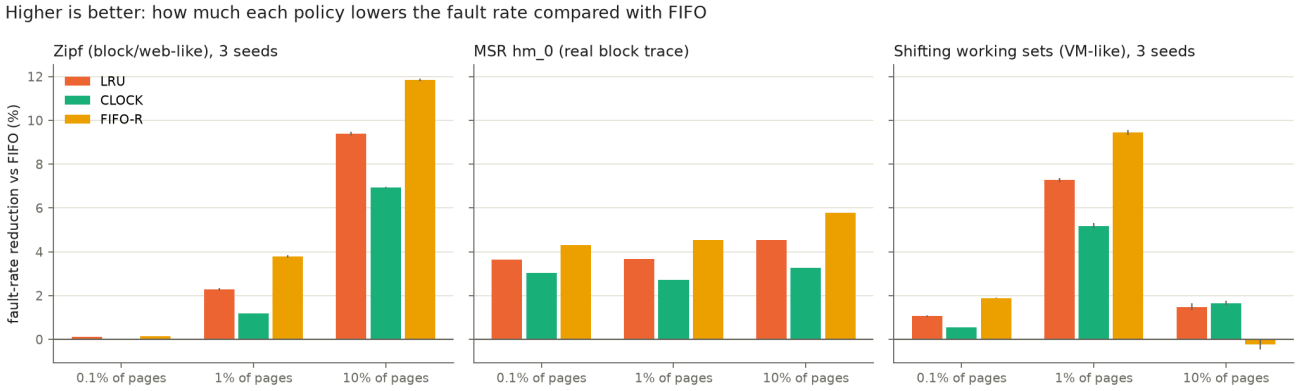


Figure 1. Fault-rate reduction compared with FIFO (higher is better). Error bars: sample standard deviation over 3 seeds (very small). MSR hm_0 is a single trace.

Workload, size	FIFO	LRU	CLOCK	FIFO-R	Best
Zipf, 1%	0.8690	0.8491	0.8588	0.8361	FIFO-R
Zipf, 10%	0.5499	0.4982	0.5118	0.4847	FIFO-R
MSR hm_0, 0.1%	0.5463	0.5263	0.5297	0.5227	FIFO-R
MSR hm_0, 10%	0.3340	0.3188	0.3231	0.3147	FIFO-R
Shift (VM-like), 1%	0.6483	0.6011	0.6148	0.5870	FIFO-R
Shift (VM-like), 10%	0.2917	0.2873	0.2868	0.2923	CLOCK ≈ LRU

Table 1. Mean fault ratio over seeds 11, 23 and 47 (sd ≤ 0.0017). The full tables, with page-ins, eviction write-backs and completion flushes, are in `results/results.md`.

Findings

- Replicated: FIFO-Reinsertion beats LRU on all Zipf runs (9 of 9) and on MSR hm_0 at every size, and FIFO is the worst policy. This matches the paper's main claim.
- New finding, one design detail decides the result: our proposal's Clock (a load sets the reference bit) loses to LRU on every Zipf and MSR run, while FIFO-R (a load leaves the bit at 0) wins. With the bit cleared, a page that is never reused is evicted on the hand's first pass. The paper calls this "quick demotion".
- Footnote 2 tested: with sudden working-set changes and a large cache (10%), FIFO-R loses its advantage. It is worse than LRU in 3 of 3 seeds and even slightly worse than FIFO, while LRU and Clock adapt better. This supports the paper's guess for VM-like workloads, but only at the large cache size. At 0.1% and 1%, FIFO-R still wins.
- Write traffic: on shift at 10%, FIFO-R has the most page-ins but the fewest eviction write-backs (9,025 vs. 9,172 for LRU). Fault counts alone do not tell the whole story.

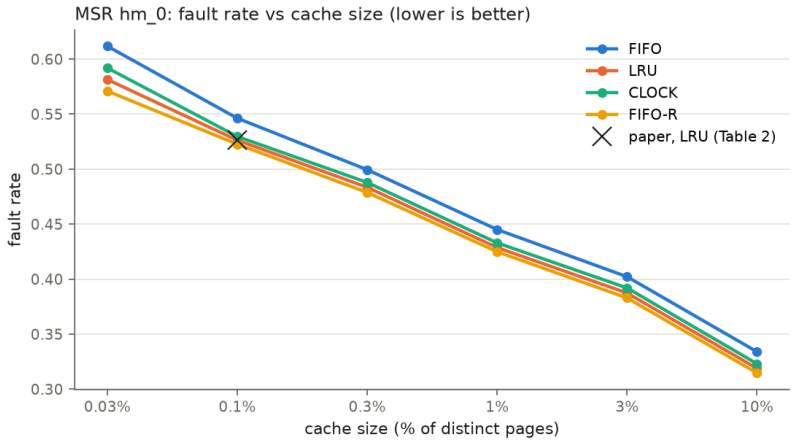


Figure 2. MSR hm_0 fault rate vs cache size. The × marks the paper's LRU value (Table 2 of [1]).

Thrashing warning (simulation heuristic)

The warning is a heuristic for this model, not a real OS thrashing detector. On the proposal's changing-locality trace (F = 4), there is no warning in phase 1. The warning turns on during the 8-page phase and clears 13 references into phase 3, before the required 20 (Fig. 3). Across middle-phase working-set sizes (Fig. 4), there is never a warning while D ≤ F (3 or 4 pages). Above that, onset is 12 references for FIFO, LRU and Clock. Because a cyclic pattern faults on every reference, the delay is set by the 20-reference window, not by the working-set size. FIFO-R is slightly later at 5 and 6 pages. It evicts newly loaded pages first, so pages 0 and 1, which were hit in phase 1, stay resident longer and give a few extra hits.

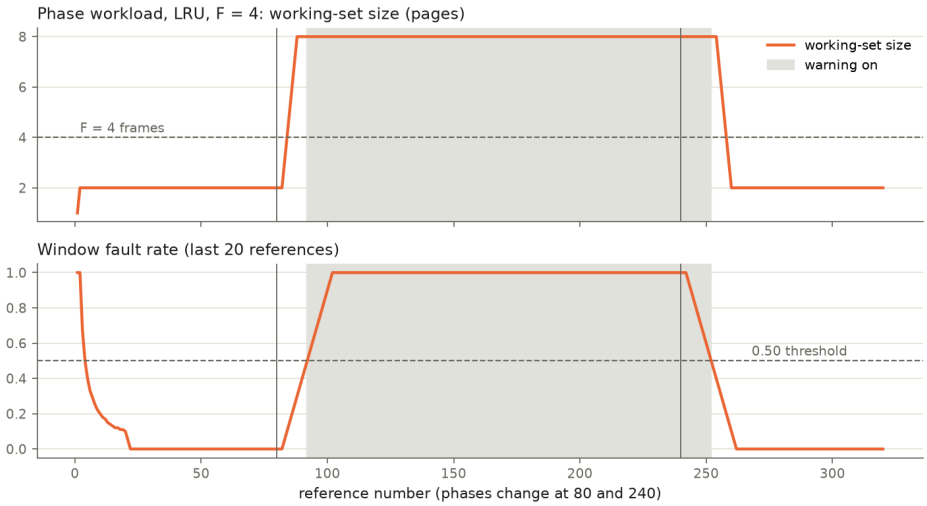


Figure 3. Working-set size and window fault rate (LRU, F = 4). Shaded: warning on.



Figure 4. References into phase 2 until the warning turns on.

5. Screenshots of the working system

```
os_project — -bash — 81×23
empty and invalid inputs (the loader's error messages below are expected)
test:1: header must be exactly 'pid,page,op', got 'pid,page'
test:1: header must be exactly 'pid,page,op', got 'page,pid,op'
test:2: expected exactly 3 fields: pid,page,op
test:2: expected exactly 3 fields: pid,page,op
test:2: pid must be a positive integer
test:2: pid must be a positive integer
test:2: pid must be a positive integer
test:2: page must be a non-negative integer
test:2: page must be a non-negative integer
test:2: op must be R or W
test:2: op must be R or W
test:3: blank line
warning rule boundaries
finished process removed from demand
changing-locality workload
    FIFO   first warning at step 92 (12 refs into phase 2)
    LRU    first warning at step 92 (12 refs into phase 2)
    CLOCK  first warning at step 92 (12 refs into phase 2)
    FIFO-R first warning at step 92 (12 refs into phase 2)
random traces, invariants after every reference

529 checks, 0 failed
```

Figure 5. End of `make test`: all 529 checks pass. The loader messages are expected, because those tests feed it invalid files on purpose.

```
os_project — -bash — 95×23
policy LRU, 439 frames, 3993316 references, 1 processes, 439187 distinct pages

   pid      refs      hits    faults   fault%  page-ins  evict-wb  flushes  first-warn
     1   3993316   1891574   2101742   52.63%   2101742         0        0           -
   all   3993316   1891574   2101742   52.63%   2101742         0        0           -

frame utilization: mean 1.000, peak 1.000
thrashing warning (heuristic): on for 0 of 3993316 steps
MacBook-Pro-8:os_project mac$
```

Figure 6. `vmsim run -p lru -f 0.1% data/msr_hm_0.csv`: LRU on the real MSR hm_0 trace gives 52.63%, the paper's Table 2 value.

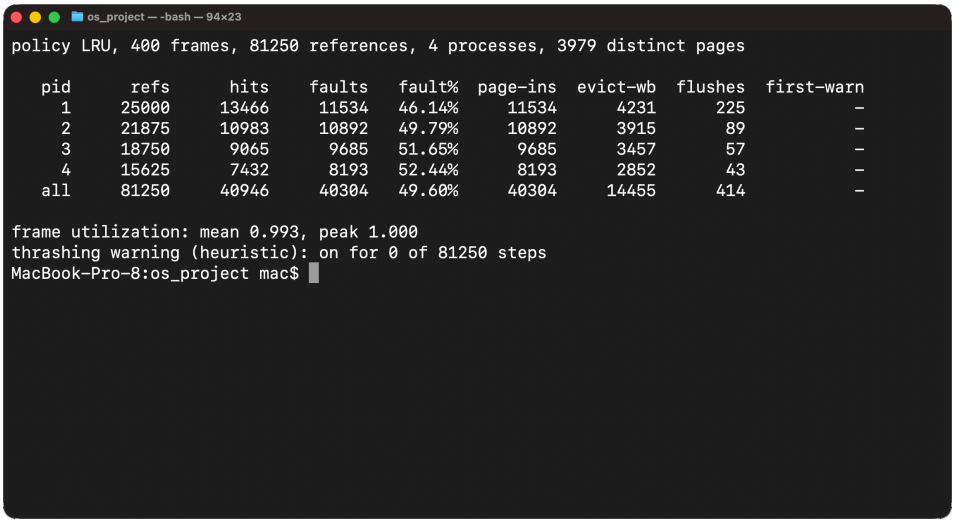


Figure 7. `vmsim run -p lru -f 400 traces/zipf_seed11.csv`: four processes sharing 400 frames, with per-process page-ins, eviction write-backs and completion flushes.

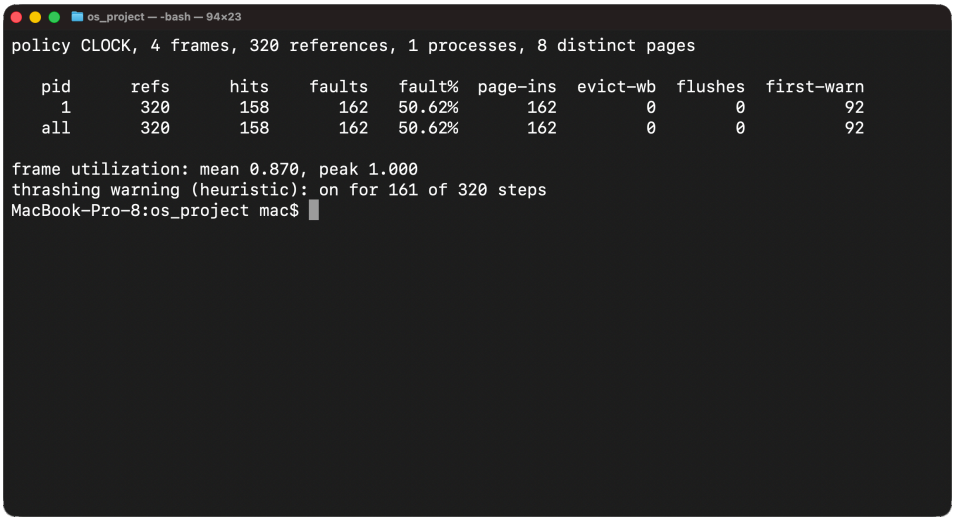


Figure 8. `vmsim run -p clock -f 4 traces/phase_ws8.csv`: on the changing-locality trace the thrashing warning first turns on at step 92.

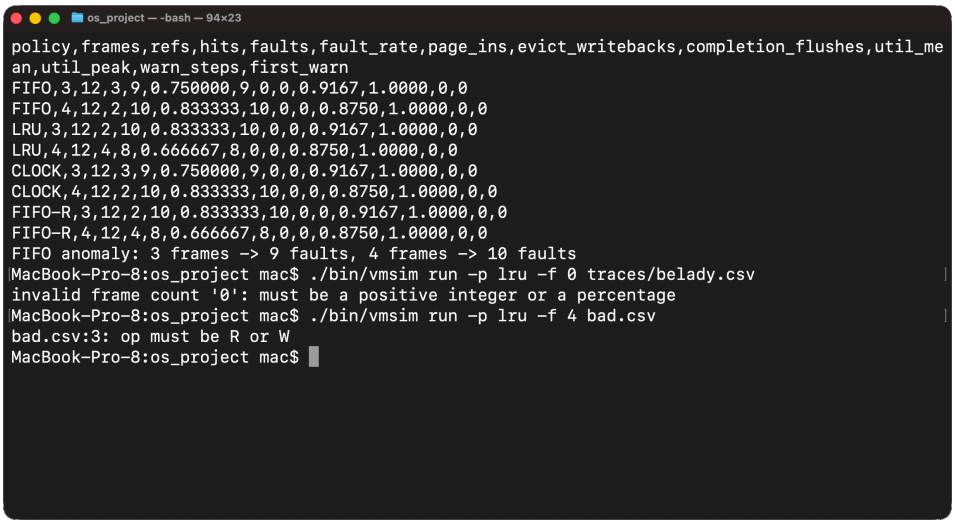


Figure 9. `vmsim compare traces/belady.csv 3,4` detects Belady's FIFO anomaly; a zero frame count and a bad trace line are rejected with clear messages.

6. Challenges faced

- Switching from Python to C. The proposal planned Python. Rewriting in C meant handling memory and data structures by hand. We kept the code small: four C files, plain arrays and one linked list.
- Scale of the real trace. MSR hm_0 has 4 million references and 439,187 distinct pages. A linear search for the oldest or least recently used page would cost up to 44,000 steps per fault, and there are more than 2 million faults. We designed for this from the start with an O(1) linked-list queue and dense page ids.
- "Clock" is not one algorithm. Our first Clock (the proposal's rule: a load sets the bit) did not reproduce the paper. Reading the paper closely showed that its FIFO-Reinsertion only keeps pages that were hit. We added it as a fourth policy and report both. This turned out to be our most interesting result.
- Pinning down rules. The hand position after inserting into a freed frame was ambiguous. We defined it as "the frame after the inserted one" and test it explicitly.
- The real trace has no read/write flag. The authors' copy of MSR hm_0 does not mark reads and writes, so that trace is read-only and has no write-backs. The synthetic traces use 30% writes.
- Smaller scale than the paper. We use 1 real trace and 3 synthetic seeds, not 5,307 traces. We replicate the direction of the result, not the paper's exact percentages.

7. Plan for completing the remaining work

Task	**Owner**
Add more MSR Cambridge traces (e.g. prn_0, proj_0, web_2, usr_2) to report the paper's "fraction of traces where FIFO-R beats LRU"	Belal Shebl
Extend the footnote-2 study: vary how often working sets shift and the cache size, and find where LRU starts to win	Muhammad Ammar Sohail
Optional: configurable window and threshold; a multi-process pressure sweep (number of processes vs. fixed F)	Muhammad Ammar Sohail
Design document and test report (expected vs. actual for every test case), code review and cleanup	Belal Shebl
Final report, slides and live demo using the saved traces	Both

References

[1] J. Yang, Z. Qiu, Y. Zhang, Y. Yue, K. V. Rashmi. FIFO can be Better than LRU: the Power of Lazy Promotion and Quick Demotion. HotOS '23, pp. 70–79, 2023. doi:10.1145/3593856.3595887

[2] D. Narayanan, A. Donnelly, A. Rowstron. MSR Cambridge traces (SNIA IOTTA trace set 388), 2007. Copy used: ftp.pdl.cmu.edu/pub/datasets/twemcacheWorkload/cacheDatasets/msr/

[3] P. J. Denning. The working set model for program behavior. Communications of the ACM 11(5):323–333, 1968.

[4] L. A. Belady. A study of replacement algorithms for a virtual-storage computer. IBM Systems Journal 5(2):78–101, 1966.